The temporal singlet as a compressed evolution
An exact memory kernel and its fluctuation seed

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Abstract

The full temporal singlet admits a finite-system isometric dilation through the adjacent charge sector. This gives its positive kernel, an exact composition defect and a closed equation with memory, without replacing unequal-time fermion anticommutators by scalars. The initial memory operator contains the previously computed dynamical singlet. A Duhamel bound proves that the ordinarily normalized bilocal tends to the identity uniformly on bounded string-time intervals in double scaling. The physical reconstruction problem therefore concerns its renormalized fluctuations and the full orthogonal memory dynamics.

1 A charge-removal isometry

Use the standard complex CAR and the neutral p-body Hamiltonian of the preceding notes. Let $\mathcal{H}_m$ be the m-fermion sector, $h = H_m$, and $H_- = H_{m-1}$. For $m > 0$, define

$$V : \mathcal{H}_m \longrightarrow \mathbb{C}^N \otimes \mathcal{H}_{m-1}, \qquad V\psi = \frac{1}{\sqrt{m}} \sum_i |i\rangle c_i \psi, \qquad K = 1_N \otimes H_-. \tag{1}$$

Since $\sum_i c_i^\dagger c_i = m$, $V^\dagger V = 1$. The unital completely positive map $\Phi(Z) = V^\dagger(1 \otimes Z)V$ is explicit; no emergent spatial coordinate has been assumed. For $S(t_a, t_b) = \sum_i c_i^\dagger(t_a) c_i(t_b)$, set

$$W(t) = e^{iKt} V e^{-iht}, \qquad B_{ab} = \frac{S_m(t_a, t_b)}{m} = W(t_a)^\dagger W(t_b). \tag{2}$$

Thus $B_{aa} = 1$, $B_{ab}^\dagger = B_{ba}$, and $\|B_{ab}\| \le 1$. For arbitrary vectors ψ_a, positivity is exact:

$$\sum_{a,b} \langle \psi_a, B_{ab} \psi_b \rangle = \left\| \sum_b W(t_b) \psi_b \right\|^2 \ge 0. \tag{3}$$

This is a concrete representation of the entire two-time object, including the same-sample correlation between neighbouring charge Hamiltonians. A generic independent-probe kernel would lose it.

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2 Composition and the full projected algebra

Let $P_b = W(t_b)W(t_b)^\dagger$. The common-middle-time product is

$$B_{ab}B_{bd} = B_{ad} - W(t_a)^\dagger(1 - P_b)W(t_d). \tag{4}$$

In particular $1 - B_{ab}B_{ba} \geq 0$. The defect describes amplitude outside the charge-removal image. For arbitrary times the product is $B_{ab}B_{cd} = W_a^\dagger W_b W_c^\dagger W_d$; subtracting the reversed product gives the exact projected commutator. This expression keeps the operator insertions that were left unresolved by the previous unprojected CAR-norm calculation.

There is also a useful energy-space expression. In the eigenbasis of h and H_- write $c^i_{\alpha a} = \langle \alpha, m-1|c_i|a, m\rangle$. For the centred bilocal $S_c(\delta) = S(-\delta/2, \delta/2)$,

$$(S_c(\delta))_{ab} = \sum_{i,\alpha} (c^i_{\alpha a})^* c^i_{\alpha b} e^{i\delta[E_\alpha - (E_a+E_b)/2]}. \tag{5}$$

Consequently its covariance is a positive Gram kernel in δ, obtained by summing $S_c(\delta)^*_{ab} S_c(\epsilon)_{ab}$ over a, b. The intermediate amplitudes must be added coherently before taking absolute squares. Their internal frequency is an adjacent-sector energy combination, not yet a quark momentum fraction or meson mass.

3 The exact memory equation

Put $\mathcal{P} = VV^\dagger$, $\mathcal{Q} = 1 - \mathcal{P}$ and

$$h_{\text{eff}} = V^\dagger K V = \left(1 - \frac{p}{m}\right) h, \qquad R = \mathcal{Q}KV, \qquad K_\perp = \mathcal{Q}K\mathcal{Q}. \tag{6}$$

The effective Hamiltonian follows from $\sum_i c_i^\dagger H c_i = (m-p)h$. Resolve $e^{iKt}V$ into its image and orthogonal components. The latter has zero initial value; integrating it and substituting back yields

$$\begin{aligned} &F(t) = V^\dagger e^{iKt} V, \qquad F(0) = 1, \\ &\dot{F}(t) = i h_{\text{eff}} F(t) - \int_0^t \mathcal{M}(t-s)F(s)\,ds, \qquad \mathcal{M}(s) = R^\dagger e^{iK_\perp s} R. \end{aligned} \tag{7}$$

This is the standard projection-memory construction [1], here applied to explicit finite matrices. No infinite-dimensional domain assumption is required. The associated Schur complement is

$$V^\dagger (z - K)^{-1} V = [z - h_{\text{eff}} - R^\dagger (z - K_\perp)^{-1} R]^{-1}, \quad \operatorname{Im} z \neq 0. \tag{8}$$

4 The existing singlet is the initial memory fluctuation

Let $A = \sum_i [H, c_i]^\dagger [H, c_i]$. The known CAR identity $\sum_i c_i^\dagger H^2 c_i = (m - 2p)h^2 + A_m$ gives

$$G := \mathcal{M}(0) = R^\dagger R = \frac{A_m}{m} - \frac{p^2}{m^2} h^2 \geq 0. \tag{9}$$

In particular $A_m \geq p^2 h^2/m$ and $[h, G] = [h, A_m]/m$. For $m = p$, $H_- = 0$, $G = 0$ and the compression is exactly autonomous. For general interacting sectors it has nonzero memory. Even the free $p = 1$ model can have $G \neq 0$: memory of a compression does not by itself diagnose interacting mesons.

At finite N, expansion of the full bilocal gives

$$S_c(\delta) = m - ip\delta h - \tfrac{1}{2}\delta^2 A_m + O(\delta^3), \qquad \frac{S_c(\delta) + S_c(-\delta)}{2} = m - \tfrac{1}{2}\delta^2 A_m + O(\delta^4). \tag{10}$$

Thus the first nonconserved even-relative-time coefficient is exactly the previously studied A, after the full flavour sum. At two centre times, the coefficient of $\delta^2\epsilon^2$ in the commutator of the even bilocals is $[A(T), A(T')]/4$. This is a local expansion, not an assertion of uniformity of the derivative expansion in p.

To connect to the existing dynamical channel, on each sector use $C_m = \mathbb{E}\,\mathrm{tr}_m\, h^2$, $a_m = \mathbb{E}\,\mathrm{tr}_m\, A_m$, and the earlier ensemble projection coefficient c_m onto $h^2 - C_m$. Define

$$Y_m = A_m - a_m 1 - c_m(h^2 - C_m 1). \tag{11}$$

Equation (9) becomes the exact decomposition

$$G = \left(\frac{a_m}{m} - \frac{p^2 C_m}{m^2}\right) 1 + \left(\frac{c_m}{m} - \frac{p^2}{m^2}\right)(h^2 - C_m 1) + \frac{Y_m}{m}. \tag{12}$$

Applying this sector by sector, $mG_{\mathrm{residual}} = Y$ is precisely the earlier singlet after its specified conserved subtraction. In the full annealed trace its DS covariance is already known:

$$\mathbb{E}\,\mathrm{tr}_0[Y(T)Y(0)] \longrightarrow W_d G_{q^2}(T), \quad W_d = 4J_s^4\left[\lambda^{-1} + \frac{4q}{1+q}\right], \quad q = e^{-4\lambda}. \tag{13}$$

This is an external centre-time correlation of the initial memory operator. It is *not* the orthogonal-time memory function $\mathcal{M}(s)$ in (7); computing that function is the next substantive problem. The distinction prevents replacing the unknown dynamics by a previously known two-point function.

5 A controlled finite-time double-scaling statement

Use the earlier Gaussian model with standard-CAR coefficient variance

$$w = 4^p \frac{J_s^2 N}{p^2} \left(\frac{p!}{N^p} \right)^2, \qquad C_m = w \binom{m}{p} \binom{N-m+p}{p}. \tag{14}$$

The sector second moment is an existing embedded-ensemble formula [2]. Taking a trace of the CAR identity gives $a_m = (2p-m)C_m + mC_{m-1}$. Since

$$\frac{C_{m-1}}{C_m} = \frac{m-p}{m} \frac{N-m+p+1}{N-m+1}, \tag{15}$$

the initial memory has the positive exact mean

$$\mathbb{E}\,\mathrm{tr}_m\, G = C_m \frac{p(m-p)(N+1)}{m^2(N-m+1)}. \tag{16}$$

For $m < p$, both h and G vanish. At half filling and $p^2/N \to \lambda > 0$, $C_m \to J_s^2/\lambda$ and

$$p\,\mathbb{E}\,\mathrm{tr}_m\, G \longrightarrow 4J_s^2. \tag{17}$$

For N even and $m = N/2$, the first limit follows explicitly from

$$C_{N/2} = \frac{J_s^2}{\lambda_N} \prod_{j=0}^{p-1} \left(1 - \frac{2j}{N}\right) \prod_{j=1}^{p} \left(1 + \frac{2j}{N}\right), \qquad \lambda_N = p^2/N. \tag{18}$$

The leading logarithms cancel; the remainder is $O(p/N + p^3/N^2)$.

More than a curvature limit is available. The intertwining defect $KV - Vh_{\mathrm{eff}} = R$ gives the exact Duhamel formula

$$e^{iKt}V - Ve^{ih_{\mathrm{eff}}t} = i \int_0^t e^{iK(t-s)} R e^{ih_{\mathrm{eff}}s}\, ds. \tag{19}$$

Using the normalized Hilbert–Schmidt norm on the source sector, unitarity and $V^\dagger$ contractivity imply

$$\|F(t) - e^{ih_{\mathrm{eff}}t}\|_{2,m}^2 \le t^2\, \mathrm{tr}_m\, G, \tag{20}$$

$$\mathbb{E}\|B_{0t} - 1\|_{2,m}^2 \le 2t^2 \left[\mathbb{E}\,\mathrm{tr}_m\, G + \frac{p^2}{m^2} C_m \right]. \tag{21}$$

The right-hand side is $O(J_s^2 t^2/p)$ at half filling. It controls the supremum over any fixed compact time interval. It does not extend to times growing like $\sqrt{p}$ or to a rescaled fluctuation field without additional estimates.

6 Full-trace limit and the reconstruction variable

The same conclusion holds for the original infinite-temperature full trace. Set $B_m = S_m/m$ for $m > 0$ and $B_0 = 1$. Charge weights are $2^{-N}\binom{N}{m}$. In the central window $|m - N/2| \le N^{2/3}$, the product expansions above are uniform:

$$C_m \to J_s^2/\lambda, \qquad p\,\mathbb{E}\,\mathrm{tr}_m\, G \to 4J_s^2, \qquad (p/m)^2 C_m = O(p^{-2}). \tag{22}$$

For example, the extra logarithmic errors are bounded by constants times $p(m - N/2)^2/N^2 + p^2|m - N/2|/N^2$, both tending to zero. Outside this window the binomial probability is at most $2e^{-2N^{1/3}}$. Uniform polynomial bounds suffice there:

$$C_m \le \frac{J_s^2}{\lambda_N}(1 + p/N)^{2p} \le \frac{J_s^2}{\lambda_N} e^{2\lambda_N}, \qquad \mathbb{E}\,\mathrm{tr}_m\, G \le (N+1)C_m. \tag{23}$$

The first inequality bounds the falling and rising products by $[m(N - m + p)]^p \le [(N + p)^2/4]^p$. Hence averaging (21) gives $\mathbb{E}\|B_{0t} - 1\|_{2,0}^2 = O(t^2/p)$. For the conventional normalized singlet,

$$\mathbb{E}\left\|\frac{2S(0,t)}{N} - 1\right\|_{2,0}^2 \le 8\mathbb{E}\|B_{0t} - 1\|_{2,0}^2 + \frac{2}{N} \longrightarrow 0. \tag{24}$$

This uses $0 \le 2Q/N \le 2$ and $\mathrm{tr}_0(2Q/N - 1)^2 = 1/N$. Centre-time conjugation preserves the norm, so the statement is uniform when $|t_b - t_a|$ is bounded.

The ordinary normalized bilocal therefore has a trivial leading limit, while the conserved-subtracted even derivative gives the nontrivial Y field in (12). A meson map must retain such fluctuations. This establishes a normalization requirement, not the complete fluctuation limit at finite relative time.

The polarization question also has useful existing guidance. Xu's tracial chord representation [3] separates field operators from auxiliary creation/annihilation operators. A nonzero bounded annihilator of a separating trace vector cannot belong to that same observable algebra. Rajgadia–Xu [4] construct state-adapted observables in a different KM-state setting. These are concrete routes to examine, but their state and Hamiltonian must be matched before transferring them to this charge-conserving model.

Validation and the next calculation

Four deterministic Hamiltonians verify twenty Gaussian-integer matrix identities, including the positive factorization

$$Z^\dagger Z = m^2 A_m - mp^2 h^2, \qquad Z = mK\mathsf{C} - (m-p)\mathsf{C}h, \qquad \mathsf{C} = \sqrt{m}V. \tag{25}$$

The same systems test the full bilocal against the dilation, positive block kernels, the composition defect, the resolvent Schur complement, 24/48-node memory integration, and twelve finite-time bounds. Five exact monomial covariance sums verify (16). Eighteen 80-digit evaluations retain finite-p deviations. For example, at $p = 256$, $J_s = 1$, the values of $p\mathbb{E}\,\mathrm{tr}_m\, G$ are 3.99867524, 3.99473404, 3.97903814 for $\lambda = 0.25, 0.5, 1$.

An independent script checks the complete four-anticommutator formula at unequal times in two interacting sectors, and reconstructs the centred bilocal from coherent charged eigenstate amplitudes. Even-relative-time difference quotients converge quadratically to A_m. The four contributions individually have squared norms much larger than their summed commutator, explicitly verifying the need to keep cancellations. These finite-system examples do not establish a nonzero DS limit of that commutator.

The next task is to calculate the covariance and spectral structure of $R^\dagger e^{iK_\perp s} R$ after the required conserved subtraction and normalization. This retains the explicit neighbouring-charge correlation and supplies an actual kernel to compare with bulk internal dynamics. Its memory parameter s, the external centre time, and the light-front quark fraction must remain distinct until a reconstruction is derived. No bulk mass has been assigned by fiat.

The dilation, Schur complement and projection-memory methods are standard mathematics. The present result is their application to the specified complex SYK bilocal; no priority claim is made. Numerical checks use ordinary floating point where indicated, not certified interval arithmetic. Analytic bounds support the finite-time limit.

References

[1] C. Widder, J. Zimmer and T. Schilling, *On the generalized Langevin equation and the Mori projection operator technique*, arXiv:2503.20457.

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[4] H. Rajgadia and J. Xu, *Emergent States and Algebras from the Double-Scaling limit of Pure States in SYK*, arXiv:2604.14387v3, Sec. 3.2.